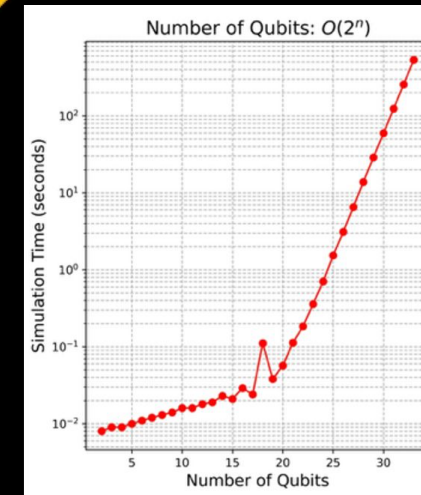
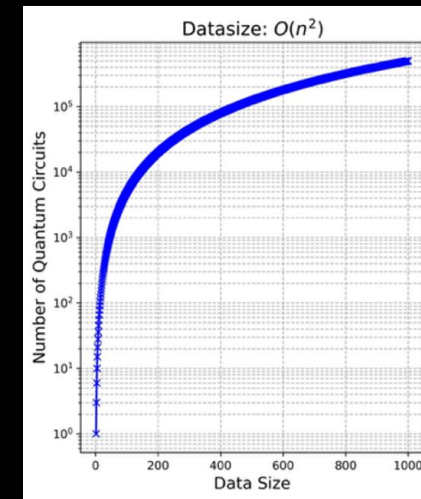
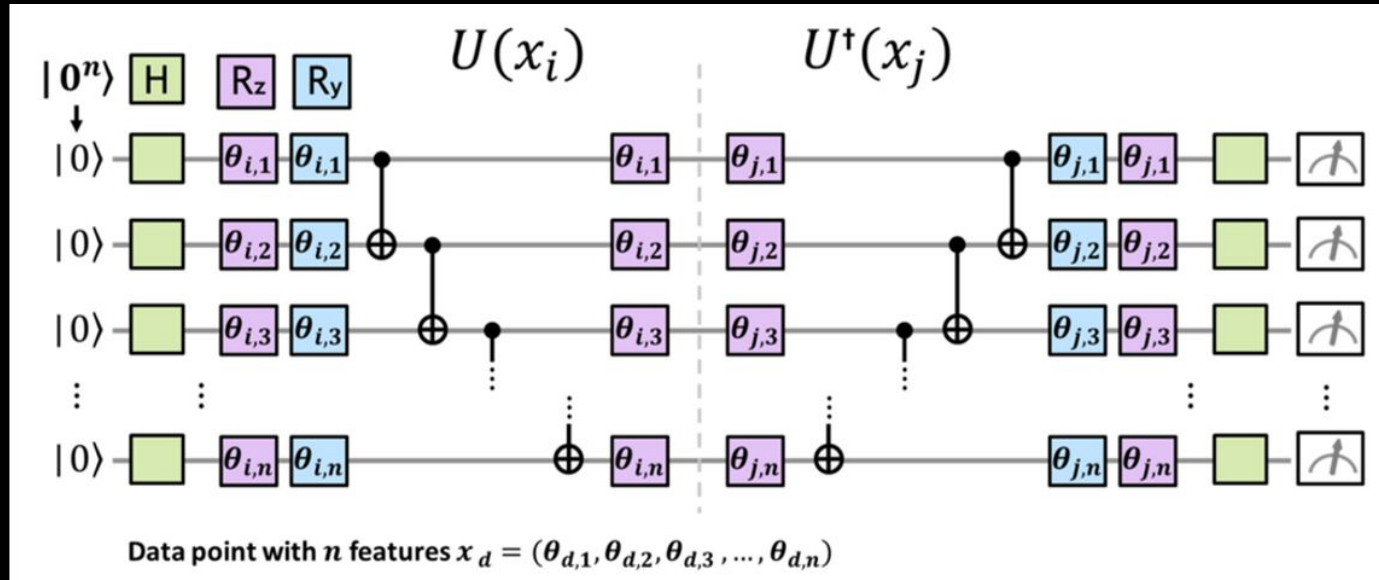
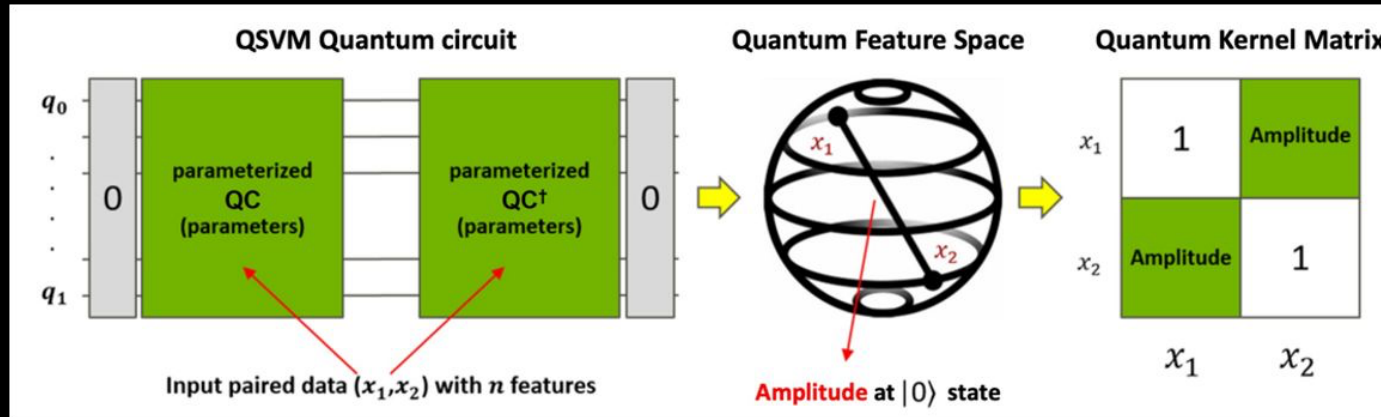


Lab 4:
Quantum-Enhanced SVM (QSVM)

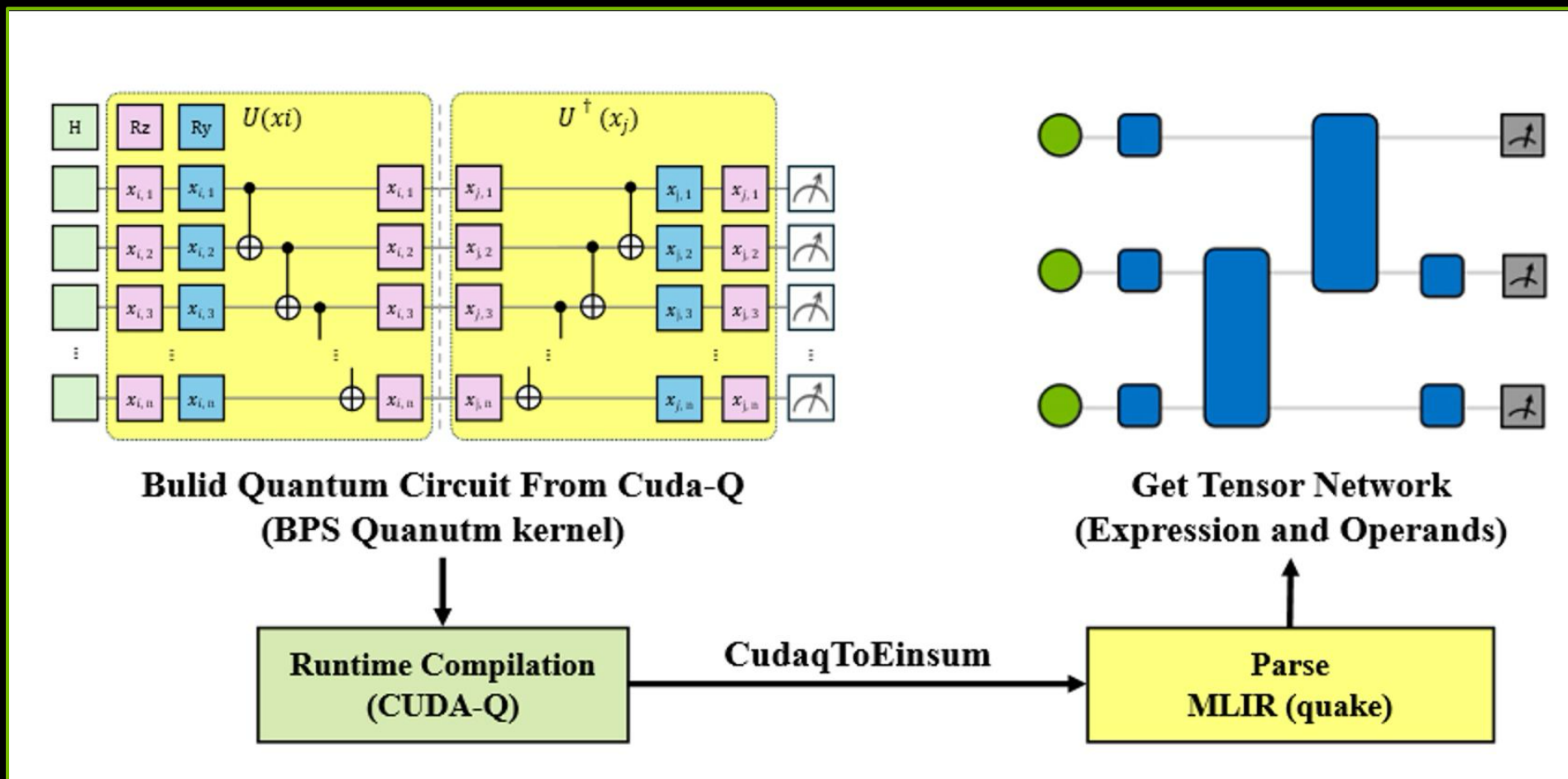
cuTN-QSVM

cuTensorNet-accelerated Quantum-enhanced Support Vector Machine



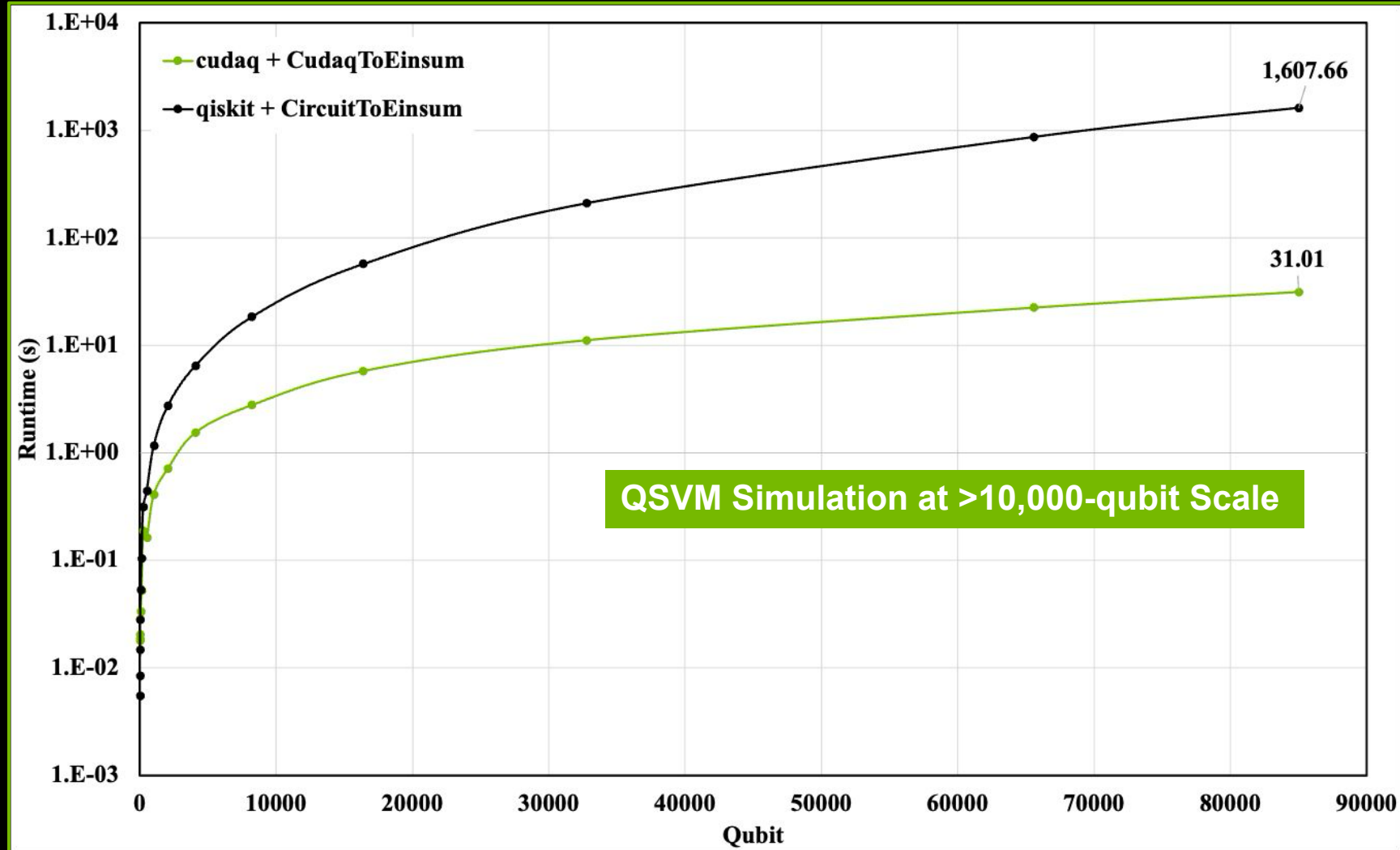
cuTN-QSVM

CudaqToEinsum: Efficient Tensor-Network Conversion from CUDA-Q



cuTN-QSVM

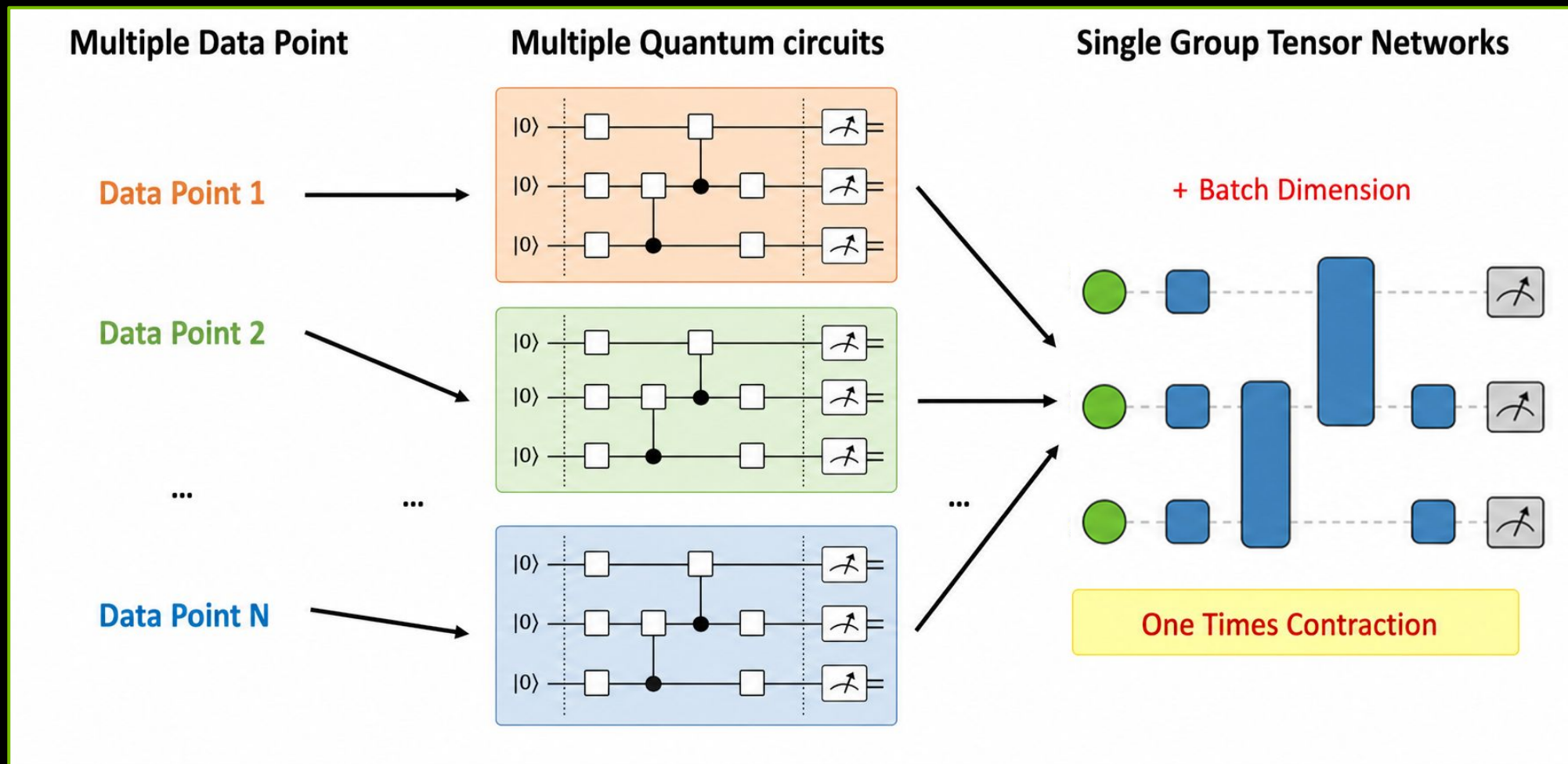
CudaqToEinsum: Efficient Tensor-Network Conversion from CUDA-Q



x50
Speed up

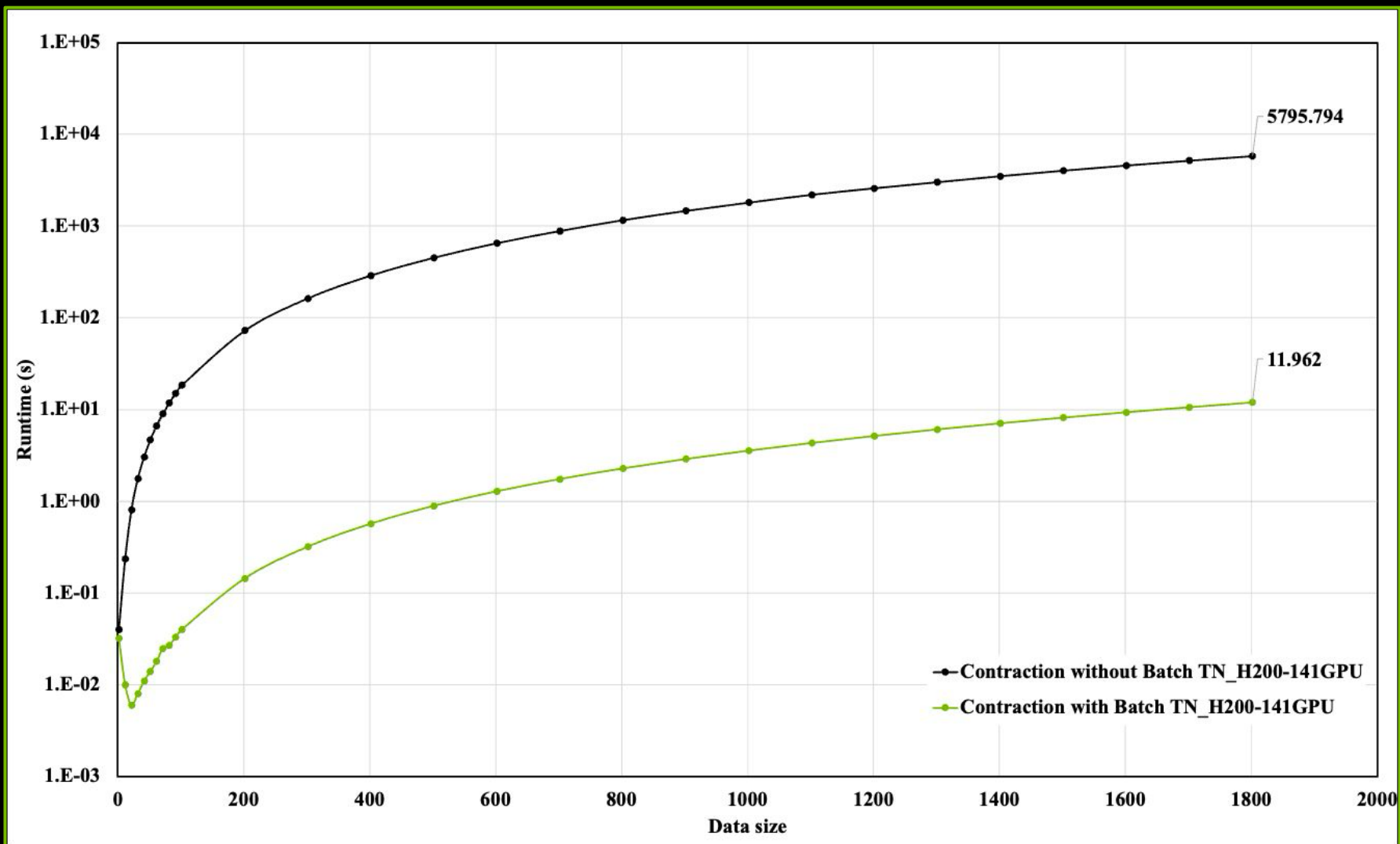
cuTN-QSVM

Efficient Batch Tensor-Network Simulation



cuTN-QSVM

Efficient Batch Tensor-Network Simulation

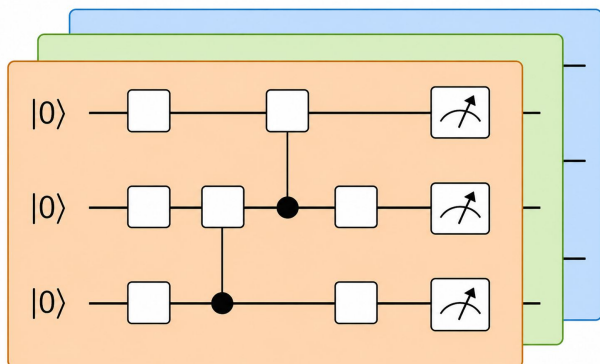


x526
Speed up

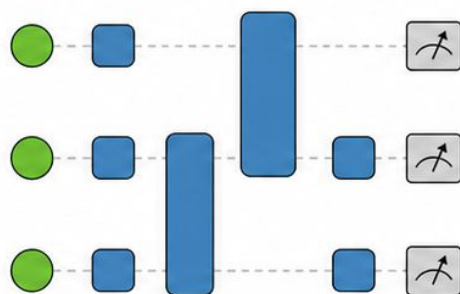
cuTN-QSVM

Multi-stream Tensor-Network Simulation Optimized by cuTENSOR

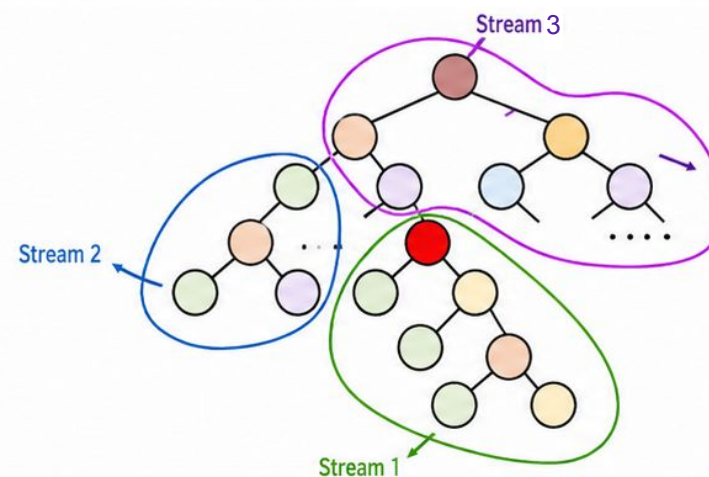
Multiple Quantum Circuits



Batch Dimension



tensor 0: abc
tensor 1: cde
tensor 2: abf
tensor 3: efg



Convert to Contraction Tree

